Recap

(S, events, P)

RV function: $S \to \mathbb{R}$

→ Discrete RV $X \in \mathcal{X} \subseteq \mathbb{R}$

PMF $p_X(x) = P(X = x)$, $x \in \mathcal{X}$

PMF properties:

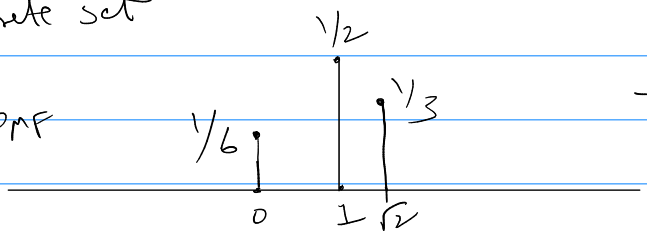
① $p_X(x) \geq 0$

$p_X : \mathbb{R} \to [0, 1]$

$\mathcal{X} = \text{supp}(p_X) = \{x : p_X(x) > 0\}$ ← discrete set

② $\sum_{x \in \mathcal{X}} p_X(x) = 1$

Σ_1: valid PMF



$S = \{a, o, b\}$ with $P(a) = 1/6$, $P(o) = 1/2$, $P(b) = 1/3$

w	$X(w)$
a	0
o	1
b	$\sqrt{2}$

Common discrete distributions

"distribution" assigns probs to subsets of $\mathbb{R}$.

→ any PMF $p_X(\cdot)$ corresponds to a distribution

Subset $Q \subseteq \mathbb{R}$

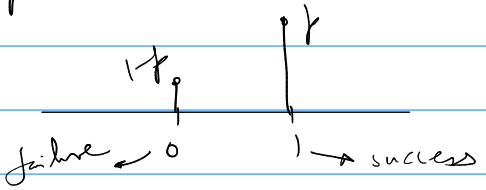
$\text{Prob}(Q) = \sum_{x \in Q \cap \text{supp}(p_X)} p_X(x)$

① Bernoulli distribution

$X \sim \text{Bernoulli}(p)$, $p \in [0, 1]$

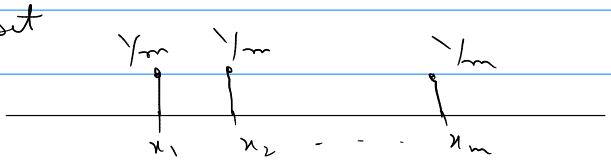
$p_X(0) = 1 - p$

$p_X(1) = p$



② Uniform on a discrete set

$X \sim \text{Uniform}(U)$, U finite set, $U = \{x_1, x_2, \ldots, x_m\} \subseteq \mathbb{R}$



③ n "indep" Bernoulli(p) $\quad X \sim \text{Binomial}(n, p)$

X = No of successes $\in \{0, 1, 2, \ldots, n\}$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\left(\frac{n}{k}\right)^k \le \binom{n}{k} \le \left(\frac{ne}{k}\right)^k$$

$$e^k > \frac{k^k}{k!}$$

$$e^{-\left(k \log \frac{1}{p} + (n-k) \log \frac{1}{1-p}\right)}$$

$$= e^{-n\left(\frac{k}{n} \log \frac{1}{p} + \left(1 - \frac{k}{n}\right) \log \frac{1}{1-p}\right)}$$

④ Geometric distribution $\quad X \sim \text{Geometric}(p)$

→ repeated "indep" Bernoulli(p) trials

Trials: 1, 2, 3, ...

X = trial of 1st success $\in \{1, 2, 3, \ldots\}$

$$p_X(k) = (1-p)^{k-1} \cdot p$$

⑤ Poisson distribution $\quad X \sim \text{Poisson}(\lambda)$

$$X \in \{0, 1, 2, 3, \ldots\}$$

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

$n \to \infty \quad np = \lambda \quad \text{Binomial}(n, p)$

$$p_{\text{Bin}(n,p)}(k) = \binom{n}{k} p^k (1-p)^{n-k} \xrightarrow[n \to \infty]{} e^{-\lambda} \frac{\lambda^k}{k!}$$